

CSA1117 - Artificial Intelligence

Descriptive written Test

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Introduction - Smart Home Management System

A smart home automation system uses AI agents to manage lighting, temperature and security by sensing environment and acting intelligently. The system must respond to real-time inputs while learning the user preferences.

1) Simple Reflex Agent

A simple reflex agent acts only based on the current sensor input.

Function in smart home.

- If motion is detected $\rightarrow$ turn on lights.
- If temperature $>$ threshold $\rightarrow$ turn on cooling.
- If door sensor triggers $\rightarrow$ send alert.

Limitations:

- No memory of past events
- cannot learn user preferences
- works only for fixed rules.

2) Model-Based Agent

This agent maintains an internal model of the environment.

Function in smart home:

- Remember previous temperature trends.

- Tracks user movement patterns.
- Understands whether a room is occupied even if motion is not detected recently.
- can adjust lighting gradually based on time of day.

Advantage:

- Handles partially observable environments and more accurate decisions than simple reflex.

3. Goal Based Agent

This agent acts to achieve specific goals.

Function in smart home:

- Goal: "Maintain comfort" - Adjust AC/heater to reach target temperature.
- Goal: "Ensure security" - lock doors automatically at night.
- Goal: "Save energy" - turn off unused appliances.

Advantages:

- Decisions are purposeful and aligned with desired outcome.

4. Utility Based Agent

Function in smart home:

- chooses optimal temperature balancing comfort and electricity cost.
- Adjusts lighting brightness based on user mood and energy consumption.
- Prioritizes security alerts based on risk level.

Advantage:

- Produces the best possible action among many choices.
- Handles trade-offs (comfort vs. cost).

Most effective approach:

A hybrid of Model based + Utility Based agent is most effective Reasoner.

- Model based memory helps understand used behaviour over time
- Utility based decision-making ensures optimal comfort, energy efficiency, and security.
- Real-Time responsiveness + learning + optimization.

UCS & IDS in Unknown Maze.

Introduction:

A robot in an unknown maze must explore and Exit without prior knowledge. Uninformed search strategies like Uniform cost search (UCS) and iterative Deepening search (IDS) help the robot systematically explore the maze.

1. Uniform cost search (UCS)

UCS expands the node with the lowest path cost.

- Robot explores paths based on actual movement cost.
- Finds the optimal (least-cost) path to the exit.
- Useful when different paths have different cost.

Advantage:

- Guarantees optimal solution and works well when costs vary.

2. Iterative Deepening search (IDS)

IDS performs repeated depth-limited searches, increasing depth each time.

- Robot explores deeper level gradually.
- Ensures complete exploration even without knowing maze size.
- Combines benefits of DFS (low memory) and BFS (completeness)

Advantage:

→ Time complexity $O(b^d)$, Space complexity: $O(bd)$ — very low

Relating to Intelligent Agents.

- (i) simple Reflex Agent - Acts only on current sensor input and cannot remember visited paths $\rightarrow$ may loop.
- (ii) Model-Based Agent. - Maintains a map of visited cells, avoids repeated explorations.
- (iii) Goal-Based Agent - Goal: "Reach exit" and choose actions that move closer to goal.
- (iv) Utility Based Agent - chooses path with best utility (shortest, safest, least cost) and works best with UCS.

Best combination:

The most effective combination is

Model-Based Agent + Uniform cost Search (UCS)

Reason:

- Model based memory helps track visited nodes and build maze map.
- UCS ensures the robot finds the optimal path to the exit.
- Handles unknown environments + cost-based decision.
- Provides both efficiency and correctness.

✶ The maze is extremely large and memory is limited, then:

Model-Based Agent + IDS is preferred due to low space usage.